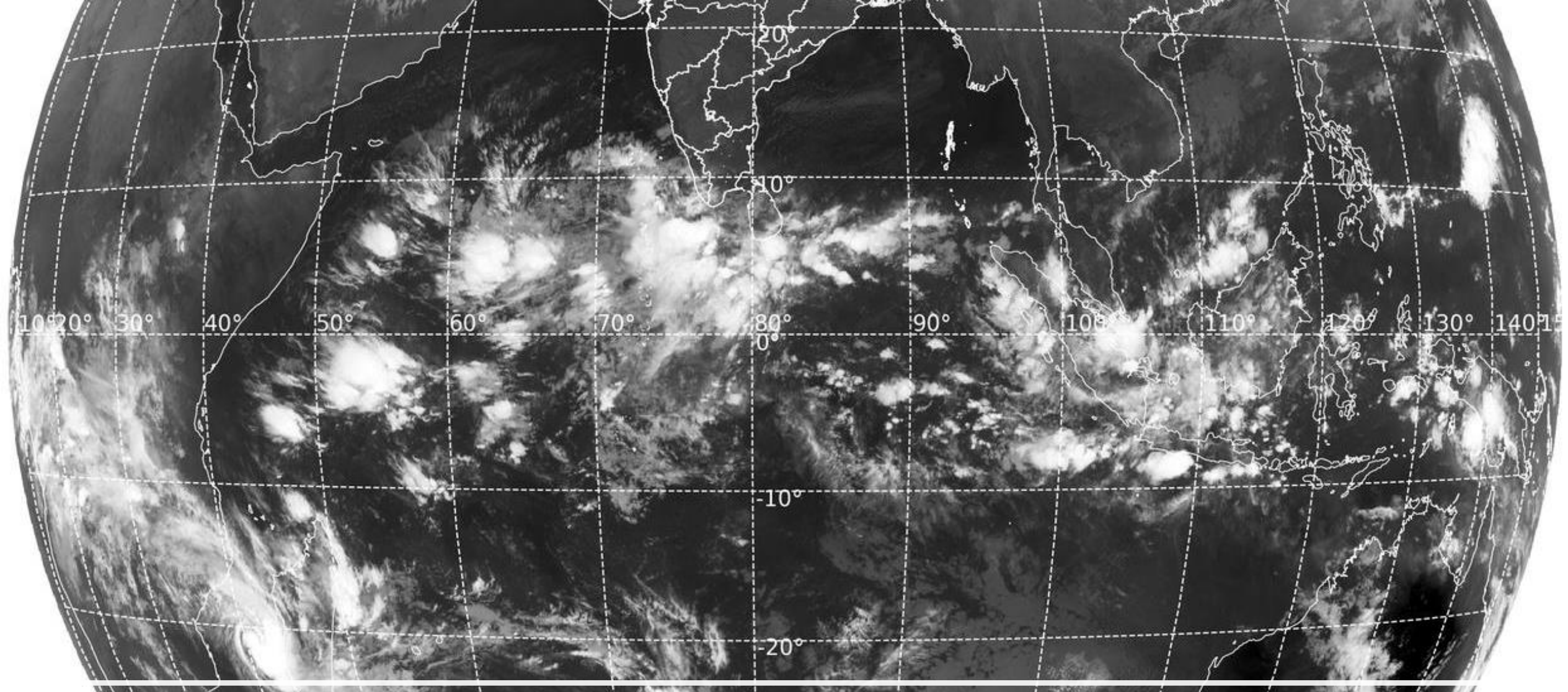




Remote Sensing and Geographic Information Science

Note: In this ppt, figures/text are adopted from various sources; this is only for class explanation purpose.

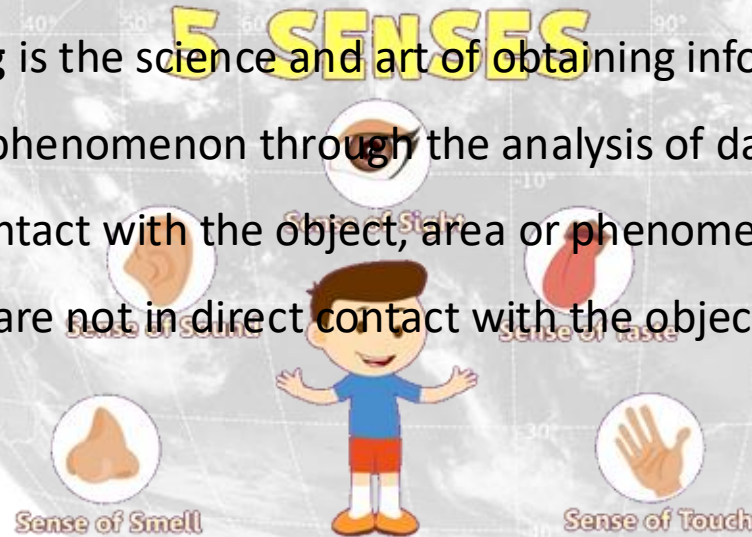
Some Brainstorming (Fun?)

- According to you what is “Remote Sensing”? Why is it called like that? Examples?



Remote Sensing is the science and art of obtaining information about an object, area or phenomenon through the analysis of data acquired by a device that is not in contact with the object, area or phenomenon under investigation.

- The sensors are not in direct contact with the objects being observed.



Some Brainstorming

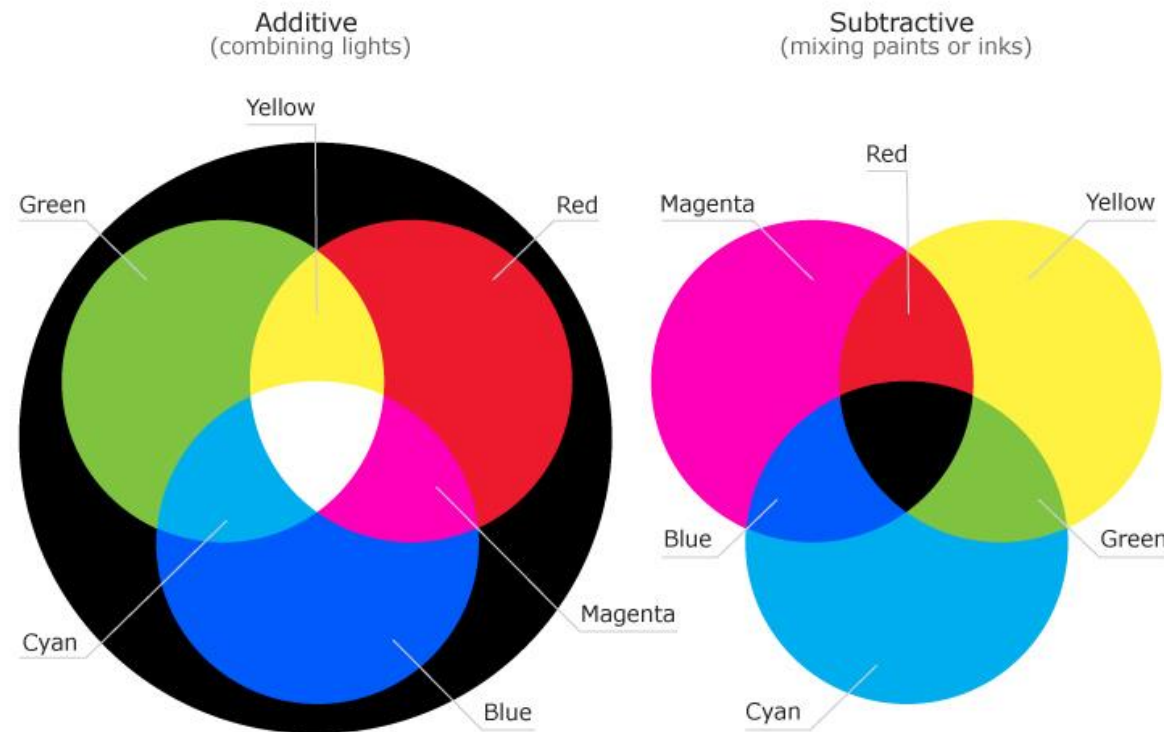
- According to you what is “Remote Sensing”? Why is it called like that? Examples?
- Can cameras be called remote sensors?
- How does the phone/DSLR cameras work?
- What is the difference between analogue and digital images?



045.292	045.292	044.616	045.292
045.292	045.292	045.292	047.320
045.292	044.616	044.616	045.968
046.644	045.968	043.264	044.616
046.644	044.616	041.912	043.264
043.940	044.616	044.616	044.616
044.616	045.292	043.940	043.940
045.968	044.616	043.940	043.264
044.616	043.264	042.588	043.264
043.264	043.940	044.616	043.940
043.940	043.940	044.616	045.292
045.968	045.292	045.968	044.616
045.292	045.968	045.968	043.940
044.616	045.292	045.292	042.588
043.264	043.940	043.940	043.264
043.264	045.968	044.616	043.264
044.616	044.616	044.616	045.292
043.940	042.588	043.940	045.292
043.940	042.588	043.264	044.616
043.940	043.940	043.940	043.940
043.940	043.264	042.588	043.940

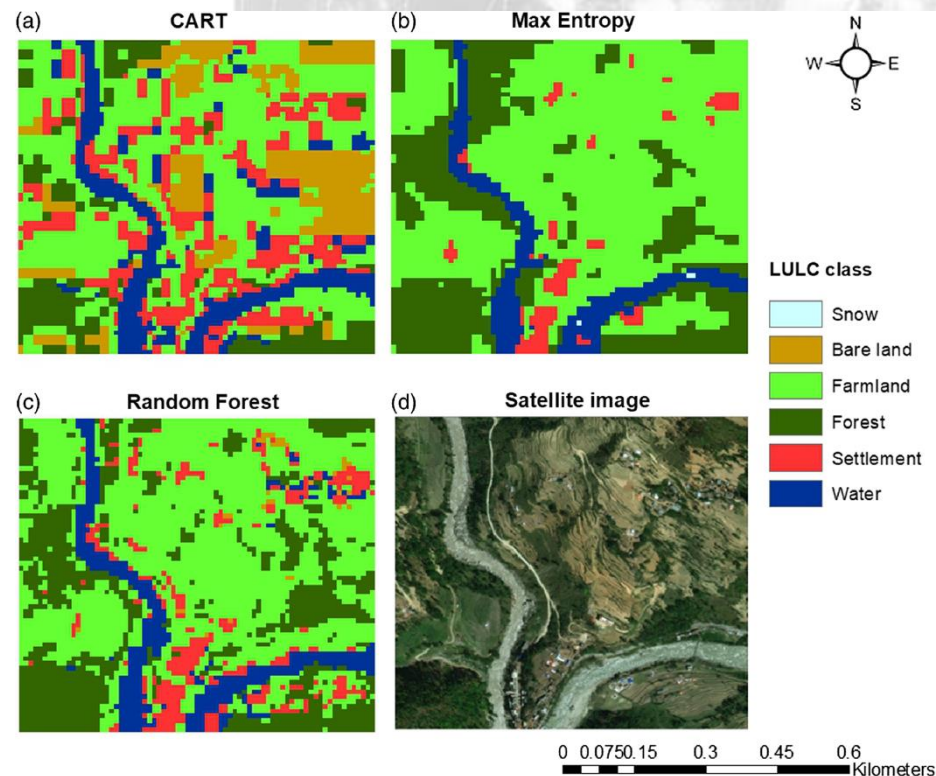
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- What is the difference between color and Black and white images?



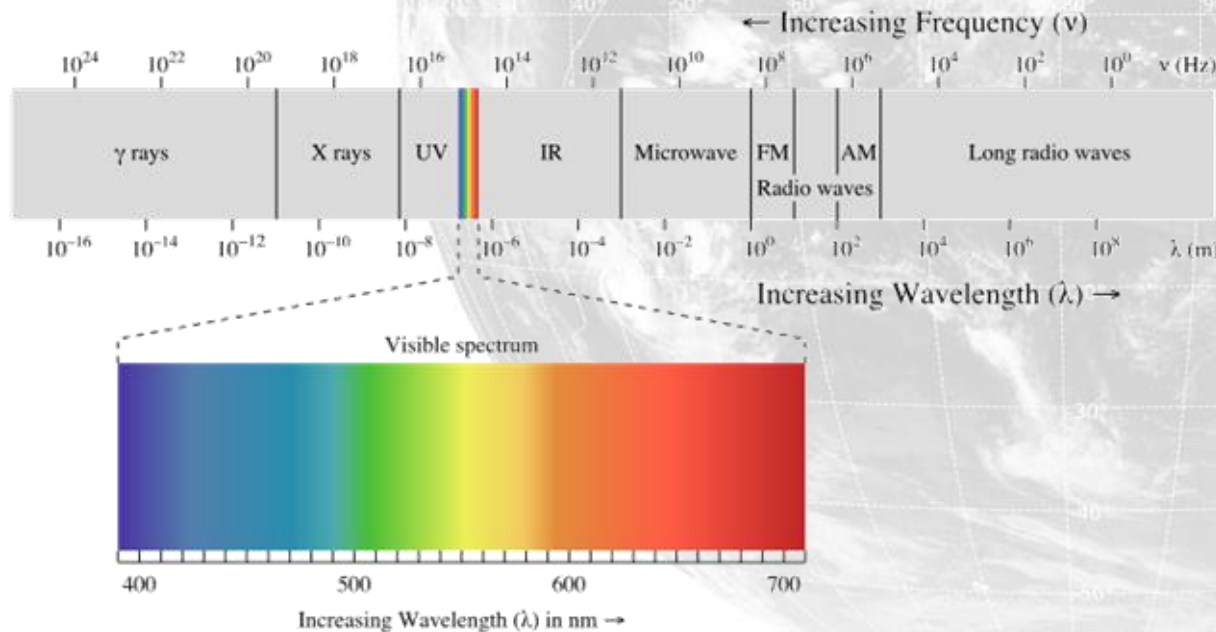
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- How does face recognition works on your phone?



Some Brainstorming

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- What is the difference between color and Black and white images?
- How does face recognition works on your phone?
- What is light? What are electromagnetic radiations (EMR)?



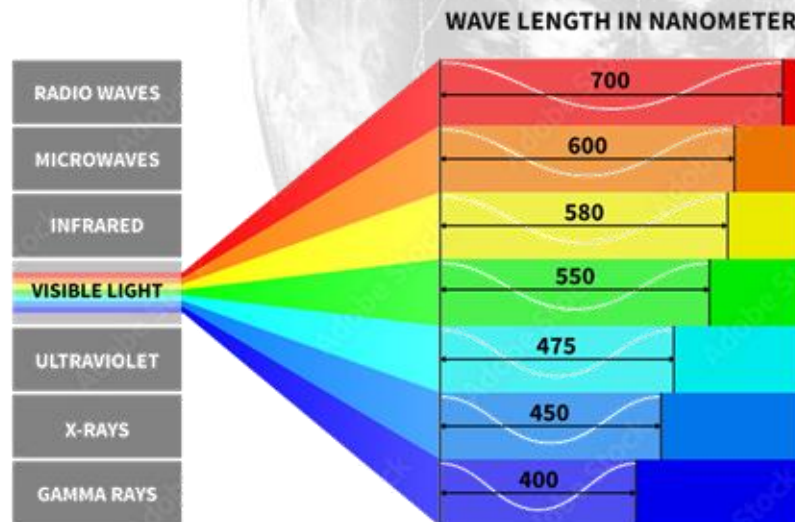
- Any object $> 0K$
- Human Eye: Visible portion – 0.4 to 0.7 μm
- Satellite sensors records EMR in a small wavelength bands called **“Spectral Channels”**

Some Brainstorming

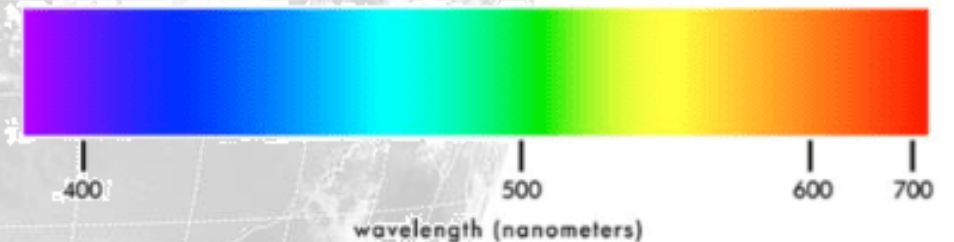
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Adobe Stock | #236854809



THE HUMAN'S VIEW



THE DOG'S VIEW



Some Brainstorming

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- What is the difference between color and Black and white images?
- How does face recognition works on your phone?
- What are electromagnetic radiations (EMR)?
- What are thermal cameras?
- How do you store and process these image datasets? How do you assign coordinates? How to link multiple datasets together? How do you prepare maps from multiple datasets? :

GIS

History of remote information sensing

- 1664-1666 Isaac Newton discovered that white light is composed of different colors.
- 1727 Johann Heinrich Schulze discovered that silver nitrate darkened upon exposure to light.
- 1814 Joseph Niepce achieves first photographic image with camera obscura - however, the image required eight hours of light exposure and later faded

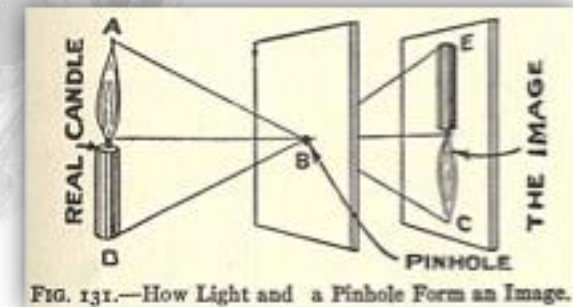


FIG. 131.—How Light and a Pinhole Form an Image.

History of remote information sensing

- 1837 Louis Daguerre's first daguerreotype - the first image that was fixed and did not fade and needed under thirty minutes of light exposure.
- Photography (as we know it) – ‘light drawing’ – invented in the early 1800’s



From the beginning, the potential of aerial photography for topographic mapping was clear

Early platforms used for aerial photos included:

- balloons
- pigeons
- kites



History of remote information sensing



A photo taken by Alfred Nobel in 1897, from a rocket



San Francisco after the earthquake and fire of 1906 (bottom photo) – kite photos.

The advantages afforded by aerial surveillance quickly become apparent during First World War.



History of remote information sensing

- After the war, planes and pilots and cameras spread widely
- Development of photogrammetric hardware (analog stereoplotters)

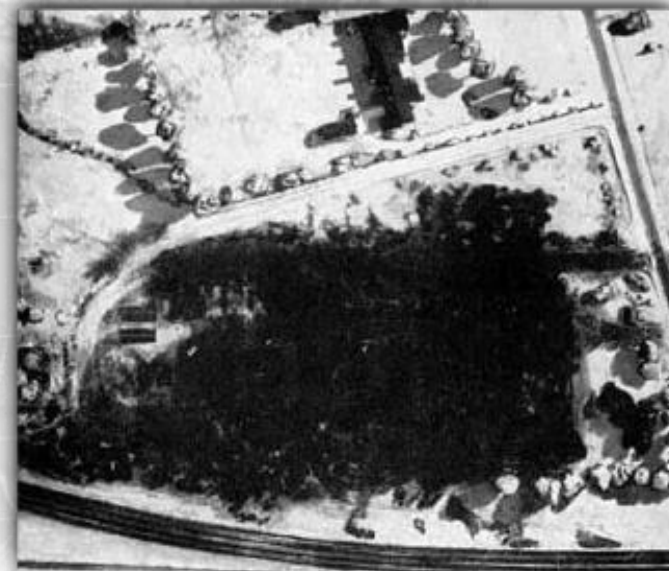


THE YEARS BETWEEN THE WARS

- Widespread use of aerial photos in forest inventory, geologic mapping, soil surveys
- Close cooperation between military and non-military groups in advancing the use of aerial photography

THE SECOND WORLD WAR (1939-1945)

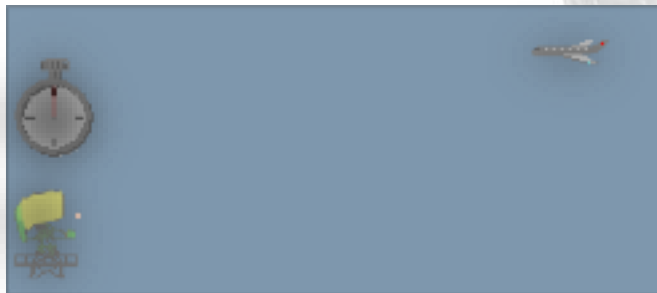
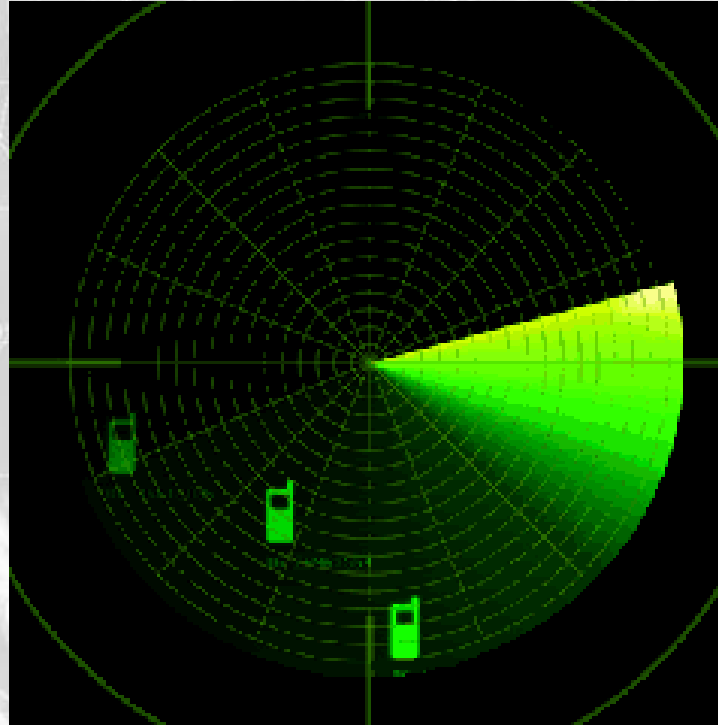
- Aerial photography emerges as a key military asset
- Photo interpretation becomes a fine art
- Photogrammetric equipment makes great strides
- Colour infrared film is developed and used for camouflage detection
- RADAR is operationalized
- Remote sensing becomes a classified technology



IR FILM: REVEALING THE HIDDEN

History of remote information sensing

RADAR: MICROWAVES, OUT OF THE OVEN



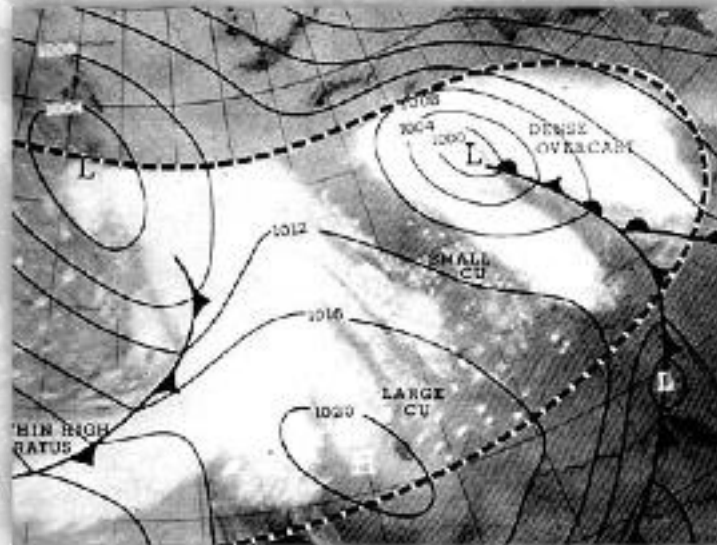
History of remote information sensing

THE COLD WAR YEARS (1945-1971)

- Military photo interpreters apply their skills to civilian topographic mapping, geology and engineering
- New platforms emerge, such as spy planes (U-2) and rockets / satellites
- Imaging systems are developed (replacing film-based devices) for radar and multispectral scanners
- The politics of remote sensing become an international issue (open skies? —nyet)

WEATHER SATELLITES

The first non-military earth observing satellite was designed to transmit television pictures back to earth showing broad weather patterns (1960).



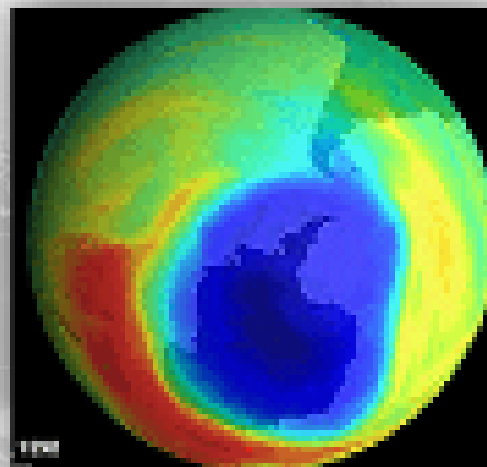
History of remote information sensing

Geographic Information Systems are developed

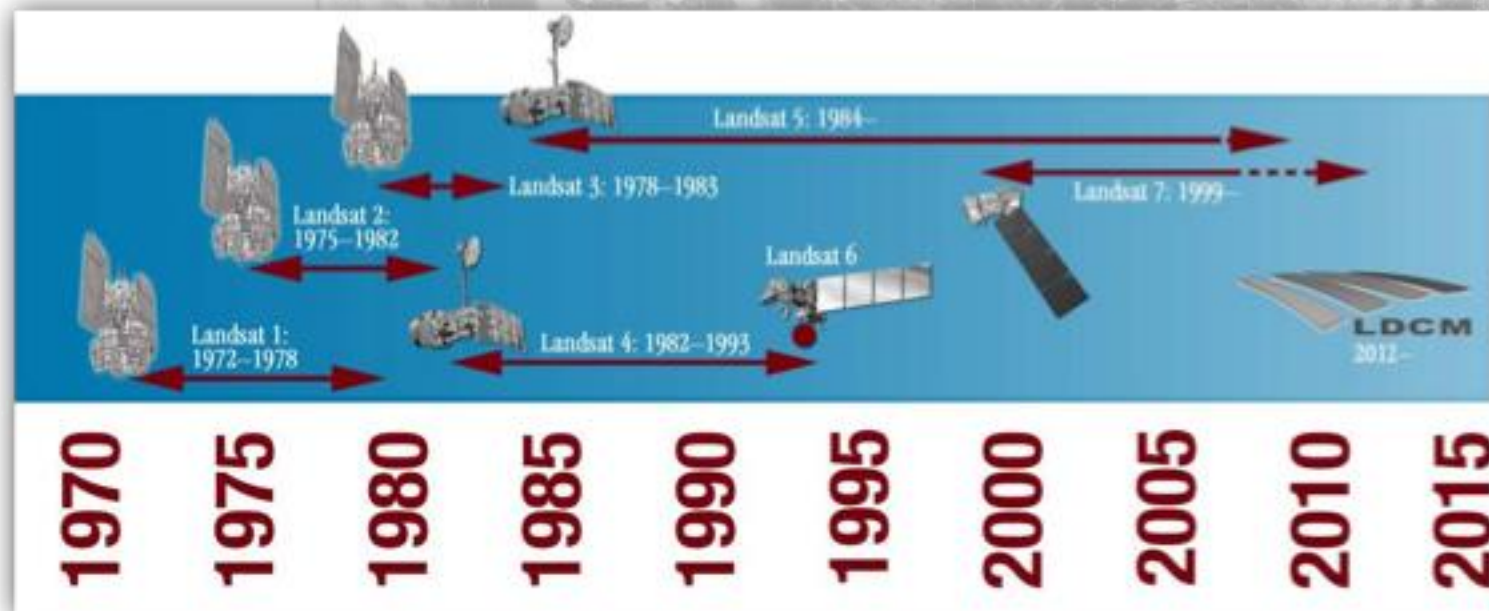
- Environment Canada's CGIS (early 60's)
- Harvard Graphics Lab's SYMAP
- US Census' DIME

Multiple satellite-based sensors launched

- LANDSAT (monitoring land resources)
- AVHRR (monitoring meteorological conditions)
- SeaSat (oceanographic research using radar)
- TOMS (ozone mapping)



Ozone 'hole' over the Antarctic



History of remote information sensing

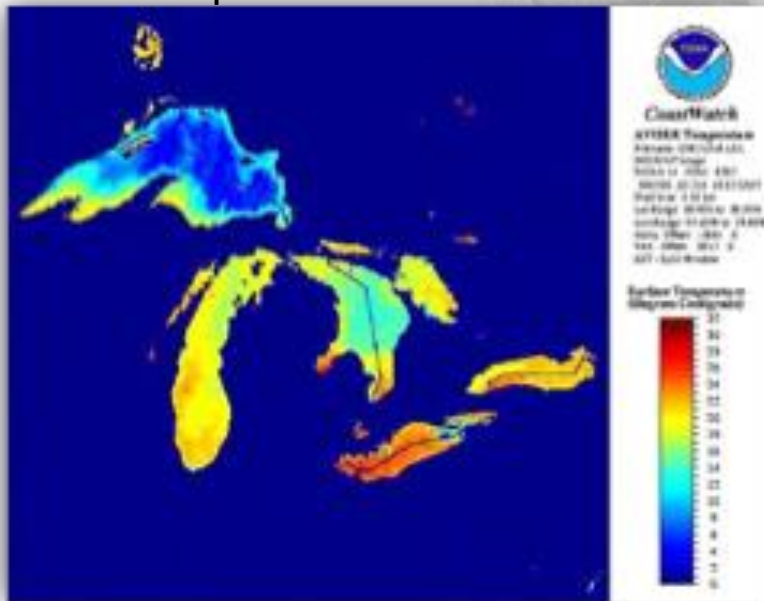
Landsat Applications

Agriculture, Forestry and Range Resources	Land Use and Mapping	Geology	Hydrology	Coastal Resources	Environmental Monitoring
Discriminating vegetative, crop and timber types	Classifying land uses	Mapping major geologic features	Determining water boundaries and surface water areas	Determining patterns and extent of turbidity	Monitoring deforestation
Measuring crop and timber acreage	Cartographic mapping and map updating	Revising geologic maps	Mapping floods and flood plain characteristics	Mapping shoreline changes	Monitoring volcanic flow activity
Precision farming land management	Categorizing land capabilities	Recognizing and classifying certain rock types	Determining area extent of snow and ice coverage	Mapping shoals, reefs and shallow areas	Mapping and monitoring water pollution
Monitoring crop and forest harvests	Monitoring urban growth	Delineating unconsolidated rocks and soils	Measuring changes and extent of glacial features	Mapping and monitoring sea ice in shipping lanes	Determining effects of natural disasters
Determining range readiness, biomass and health	Aiding regional planning	Mapping volcanic surface deposits	Measuring turbidity and sediment patterns	Tracking beach erosion and flooding	Assessing drought impact
Determining soil conditions and associations	Mapping transportation networks	Mapping geologic landforms	Delineating irrigated fields	Monitoring coral reef health	Tracking oil spills

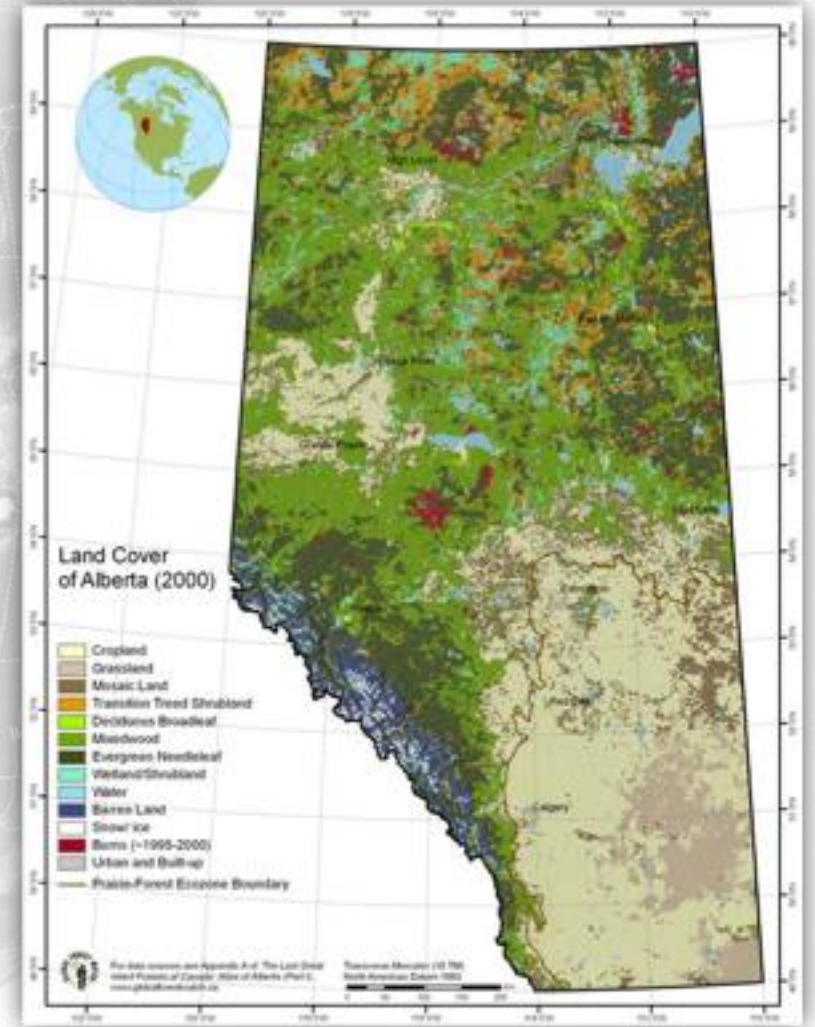
History of remote information sensing



Weather patterns over North America

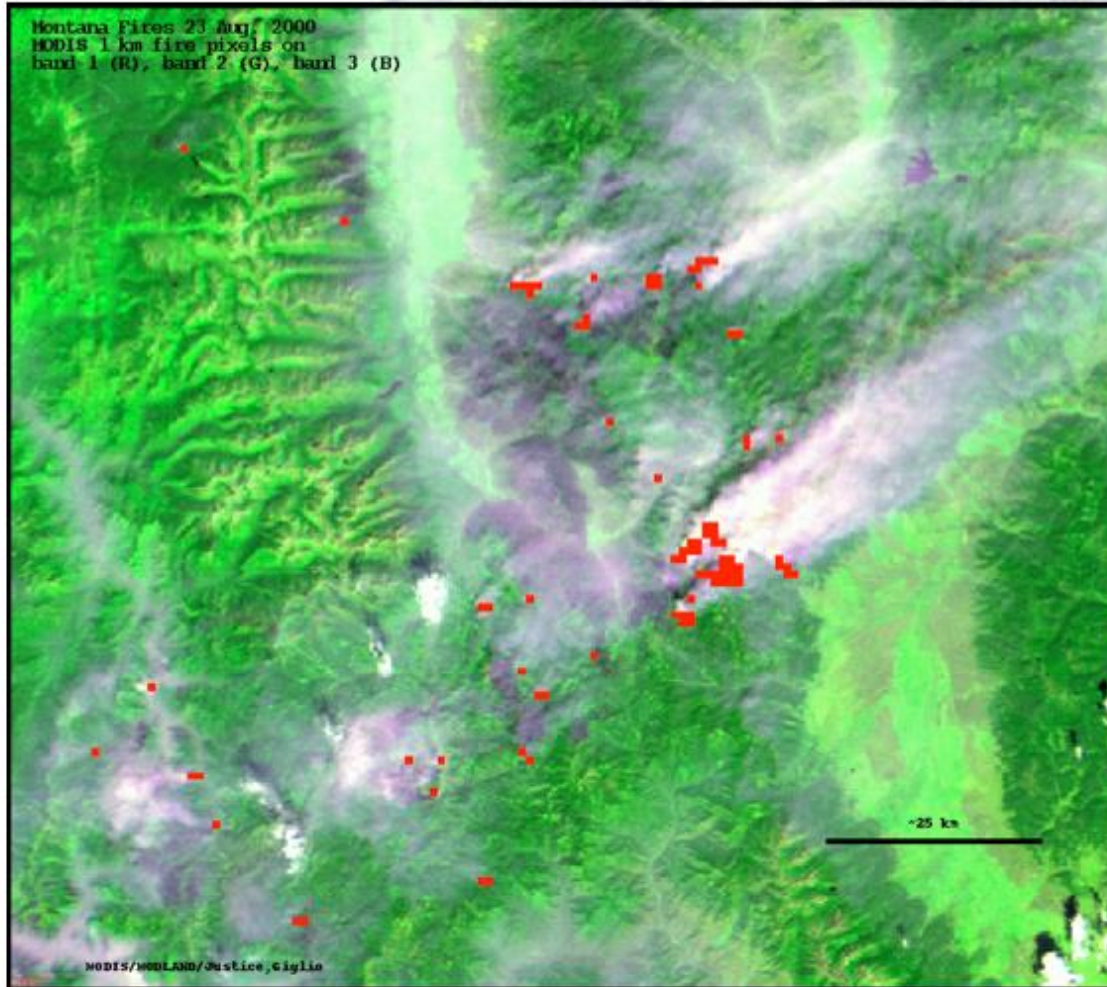


The temperature of the Great Lakes in fall.



Land cover mapping

Montana Fires, August 23 2000



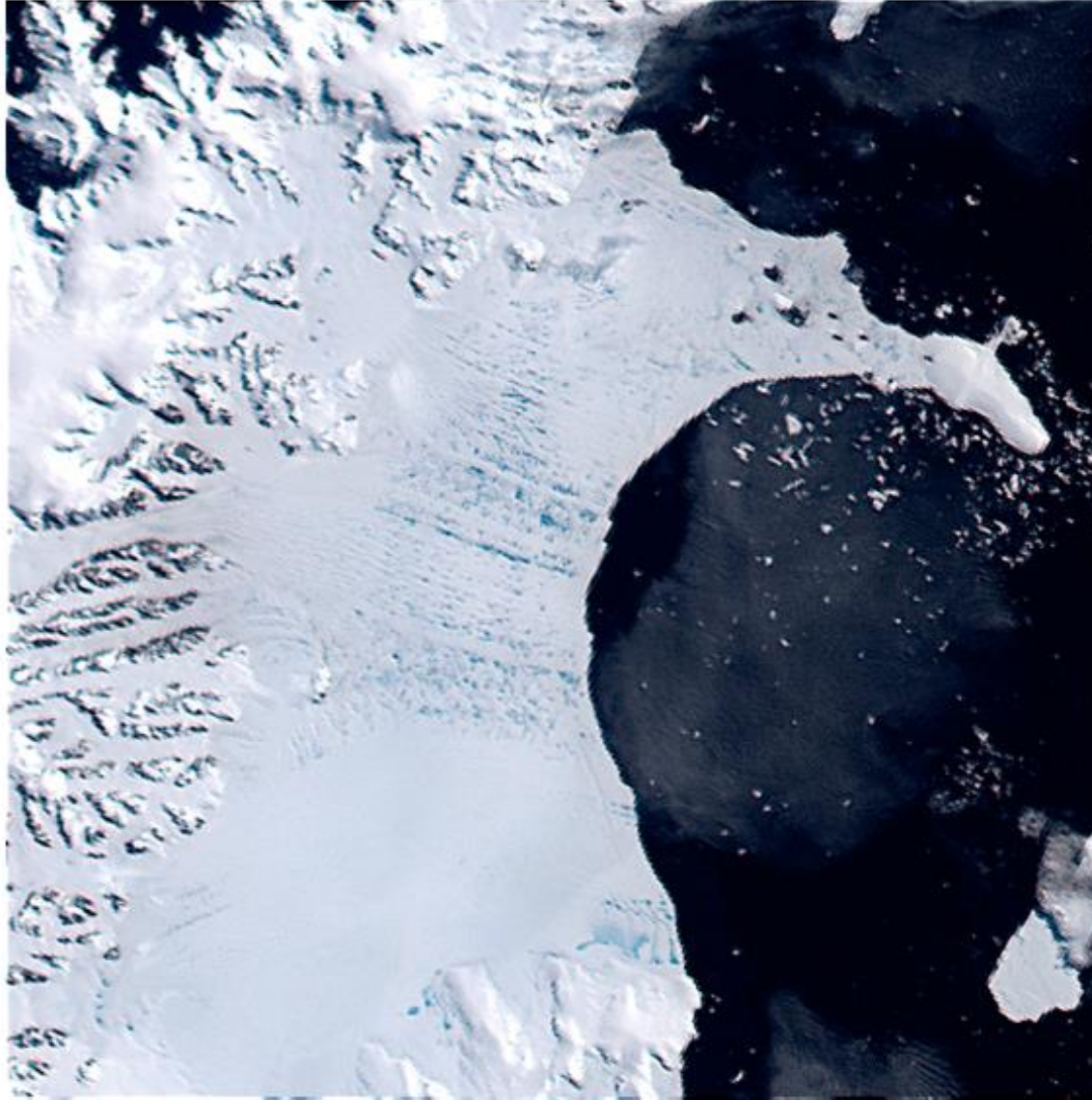
Fires superimposed on MODIS 250m surface reflectance product

MODIS thermal bands are used to map the location and strength of active fires, and burned area extent.

Vegetation maps also provide information about fire dynamics

Smoke plumes can be tracked for health monitoring

Breakup of the Larsen B Ice Shelf



MODIS
imagery from
January 31, 2002-
March 6, 2002

Courtesy of Ted Scambos, NSIDC

Arches National Park

Jem: Moab Tongue of Entrada

T_{Rw}: Wingate Sandstone

JT_{Rn}: Navajo Sandstone

Qa2: Alluvium

Qeas: Sand Deposits

Je: Slickrock Member of Entrada

Jem: Moab Tongue of Entrada

Jmt: Tidwell Member of Morrison

"Spectro-lithologic" Map

From AVIRIS 1991 Data

Clark et al, USGS

Geologic Mapping with AVIRIS

Kmu: Upper Mancos Shale

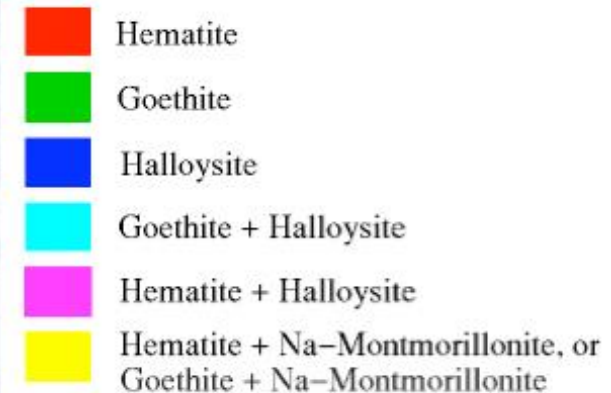
Kmf: Ferron Sandstone Member of Mancos Shale

Kml: Lower Mancos Shale

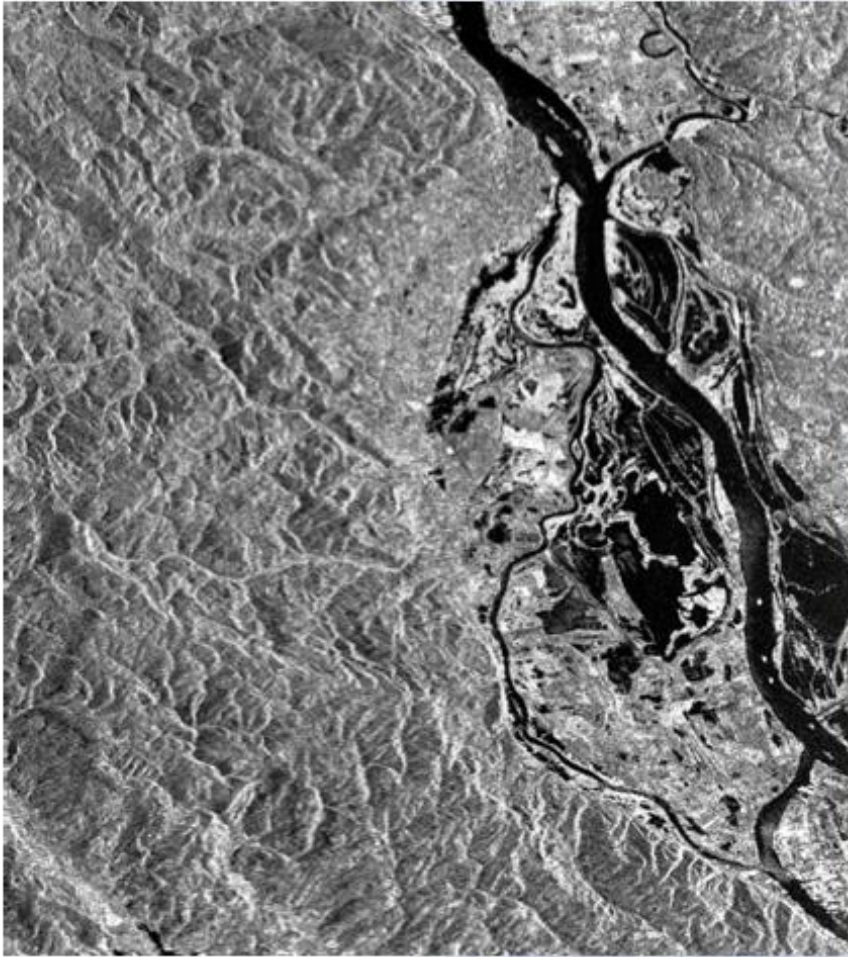
Kcm: Cedar Mountain

Jmbb: Brushy Basin Member of Morrison

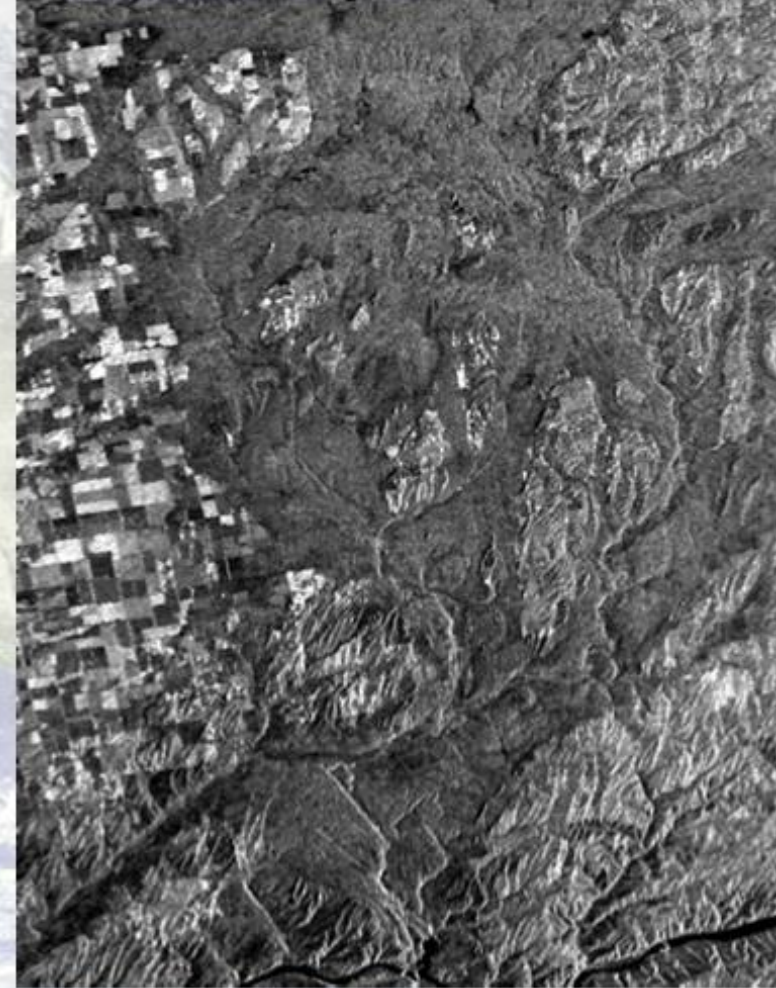
Jmsw: Salt Wash Member of Morrison



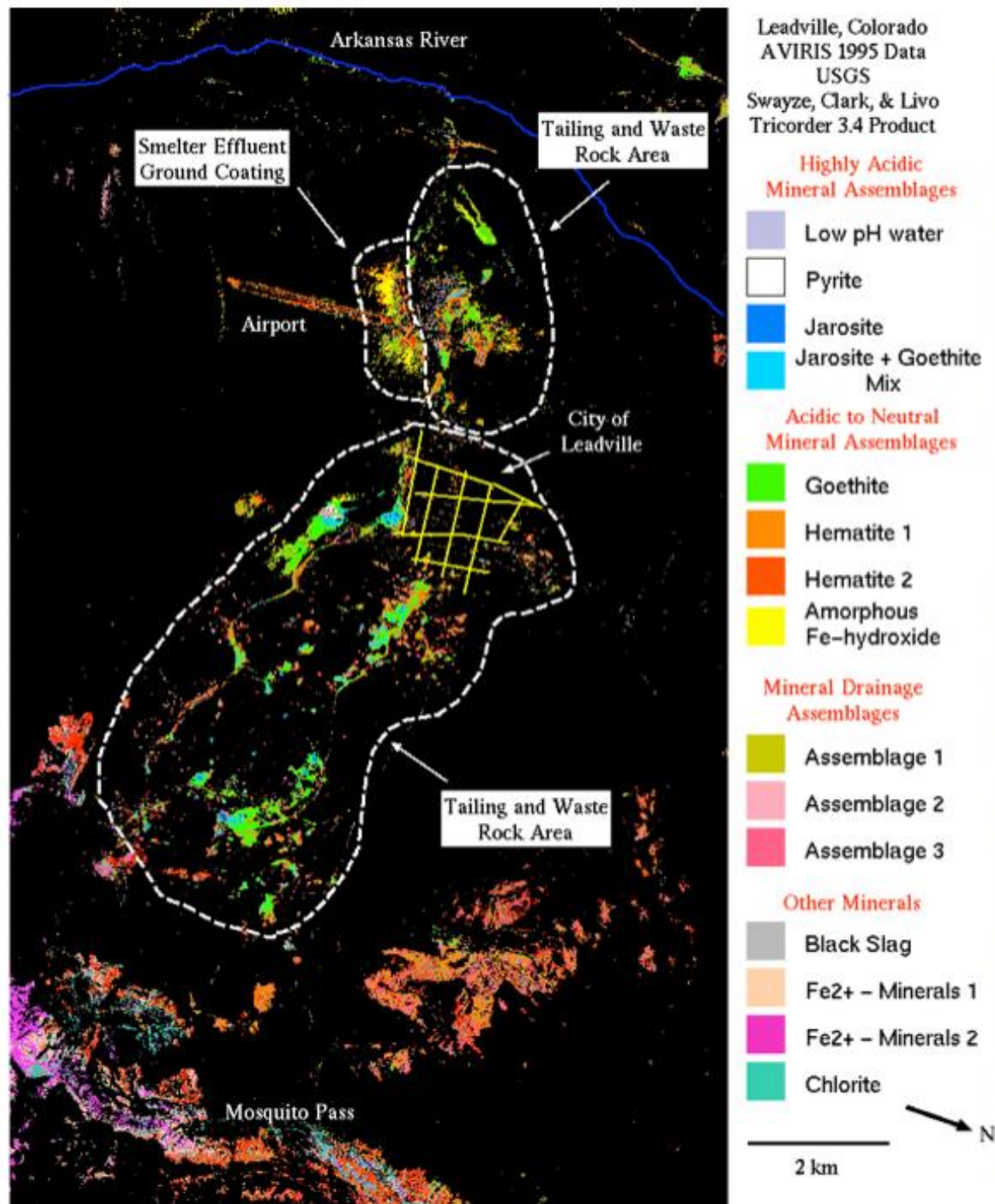
Mapping the 1996 Oregon Floods



RADARSAT image 13 Feb
Columbia River near Portland
just past peak flood stage.



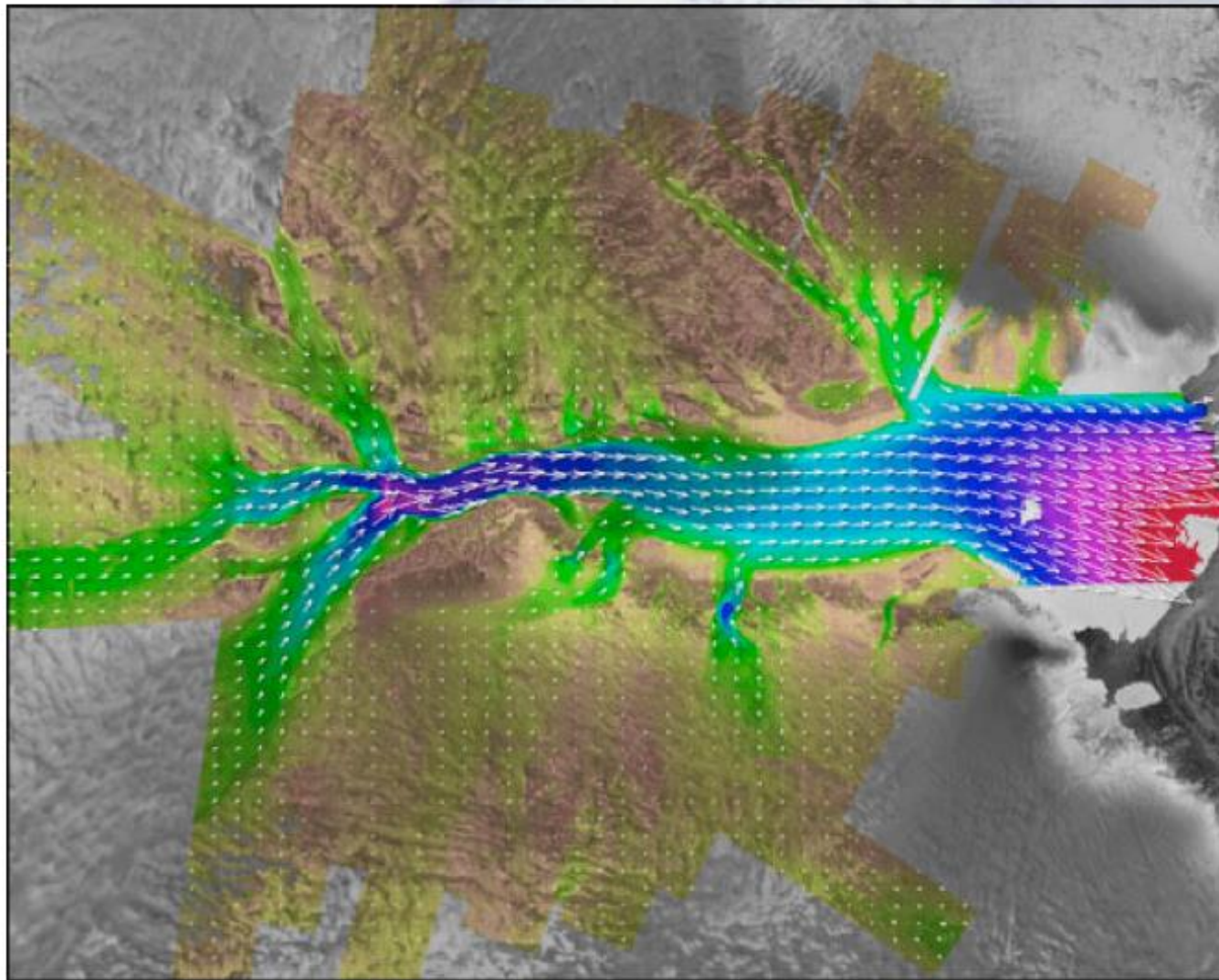
RADARSAT image 14 Feb
showing flood damaged areas



Mining Waste in Leadville, CO

AVIRIS image showing distribution and types of acid-generating minerals

Lambert Glacier, Antarctica



Interferometric
Synthetic
Aperature
Radar (InSAR)
uses phase
difference
between two
images to
determine
movement

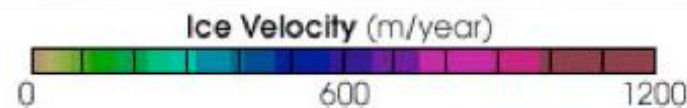


Image courtesy CSA, NASA, Ohio State Univ., JPL, ASF

Wetlands of the Gulf Coast

Multispectral and
multiangular views
of coastal wetlands off
the coast of Louisiana



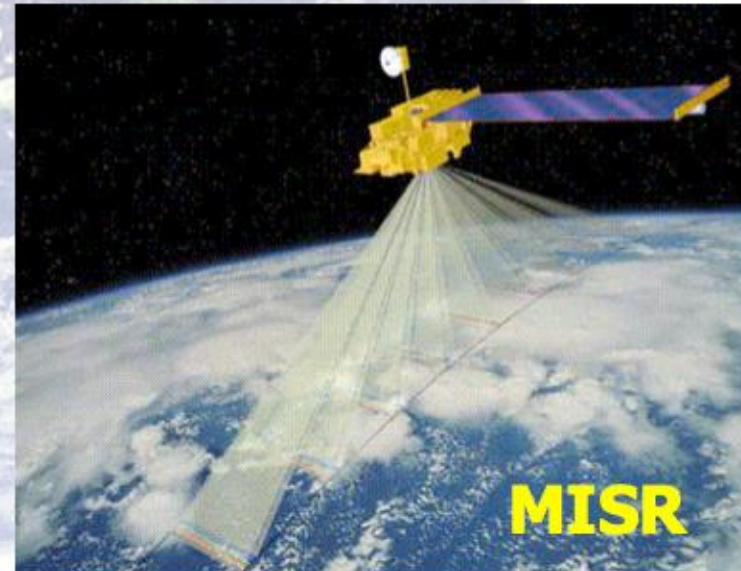
True Color



False Color (near IR=red, red=green, green=blue)

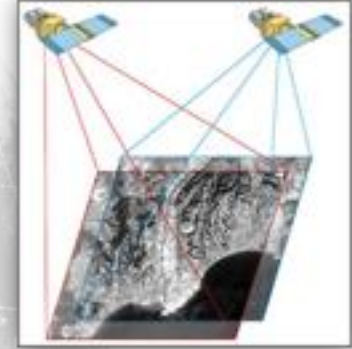


False Color (red=60° aft camera red, green=nadir camera green, blue=nadir camera blue)



History of remote information sensing

- France launches the first non-US, non-Russian satellite in 1986 (SPOT) providing the highest resolution, commercially-available imagery, at the time **(10 m panchromatic)**
- SPOT allowed for off-nadir viewing, enabling stereoscopic coverage to be collected
- GIS use explodes
- GPS developed



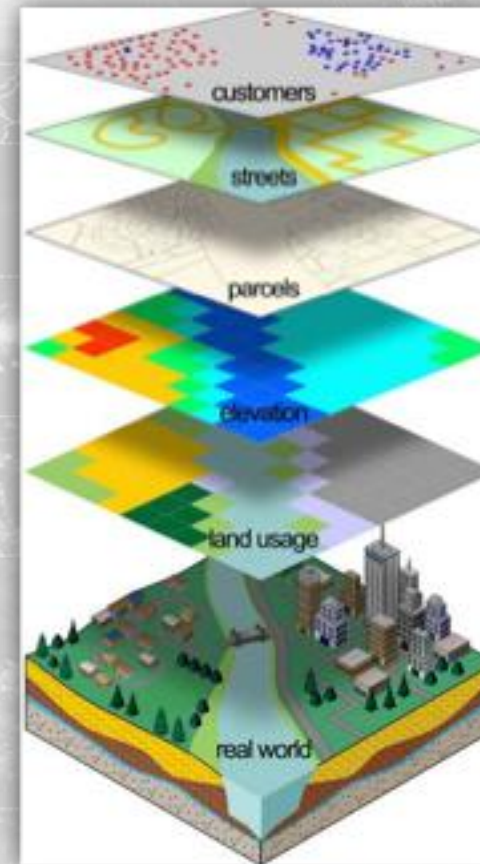
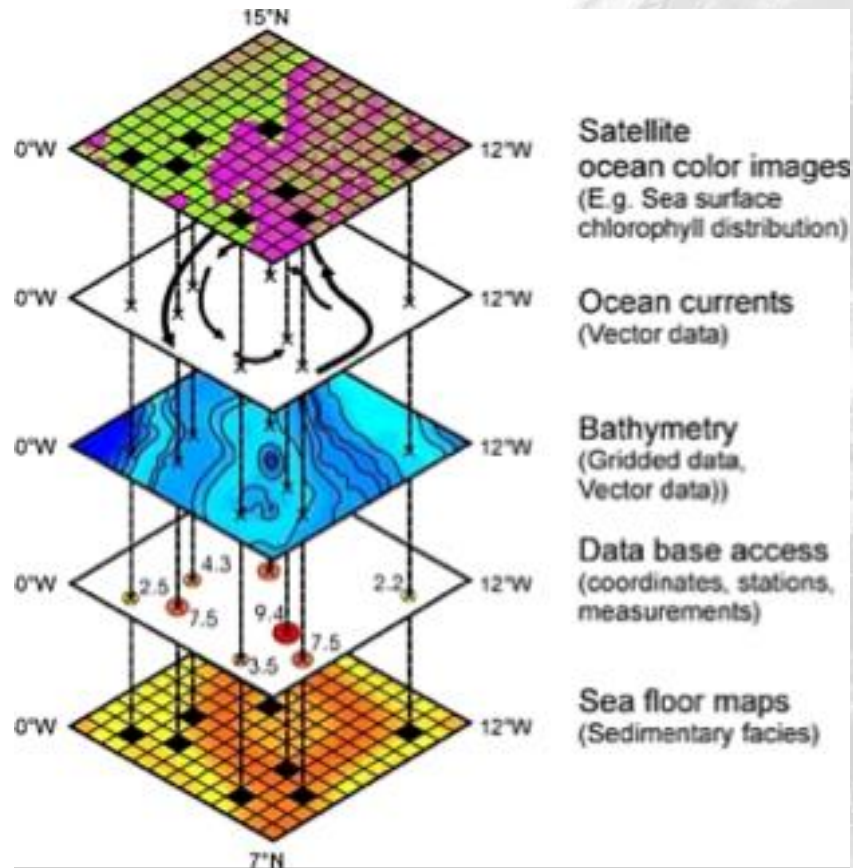
GEOLOCATION FOR ALL (AKA GPS)

- Originally developed for the military to enable locational awareness
- Originally the signal was 'fuzzified' so that civilian users couldn't get accurate results



- Ever-higher image resolutions -
- GPS in every cell phone, car, backpack
- All appliances linked to the web, RFIDs everywhere

GIS integrates disciplines and technologies such as remote sensing, surveying, photogrammetry, spatial analysis, cartography, computer science.



WHAT IS A GEOGRAPHIC PHENOMENON?

*"Almost everything
that happens,
happens somewhere.
Knowing where
something happens is
critically important."
[Longley et al., 2011]*

A geographic phenomenon is something that can:

- be named or described;
- be georeferenced (located);
- be assigned a time (interval) at which it is/was present.

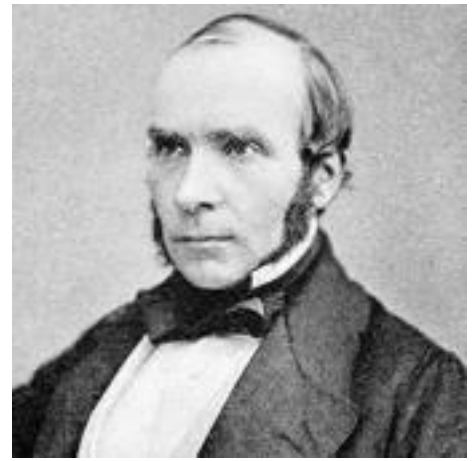
FIRST SPATIAL ANALYSIS

In 1854

During the mid-19th century, London suffered a series of cholera outbreaks.

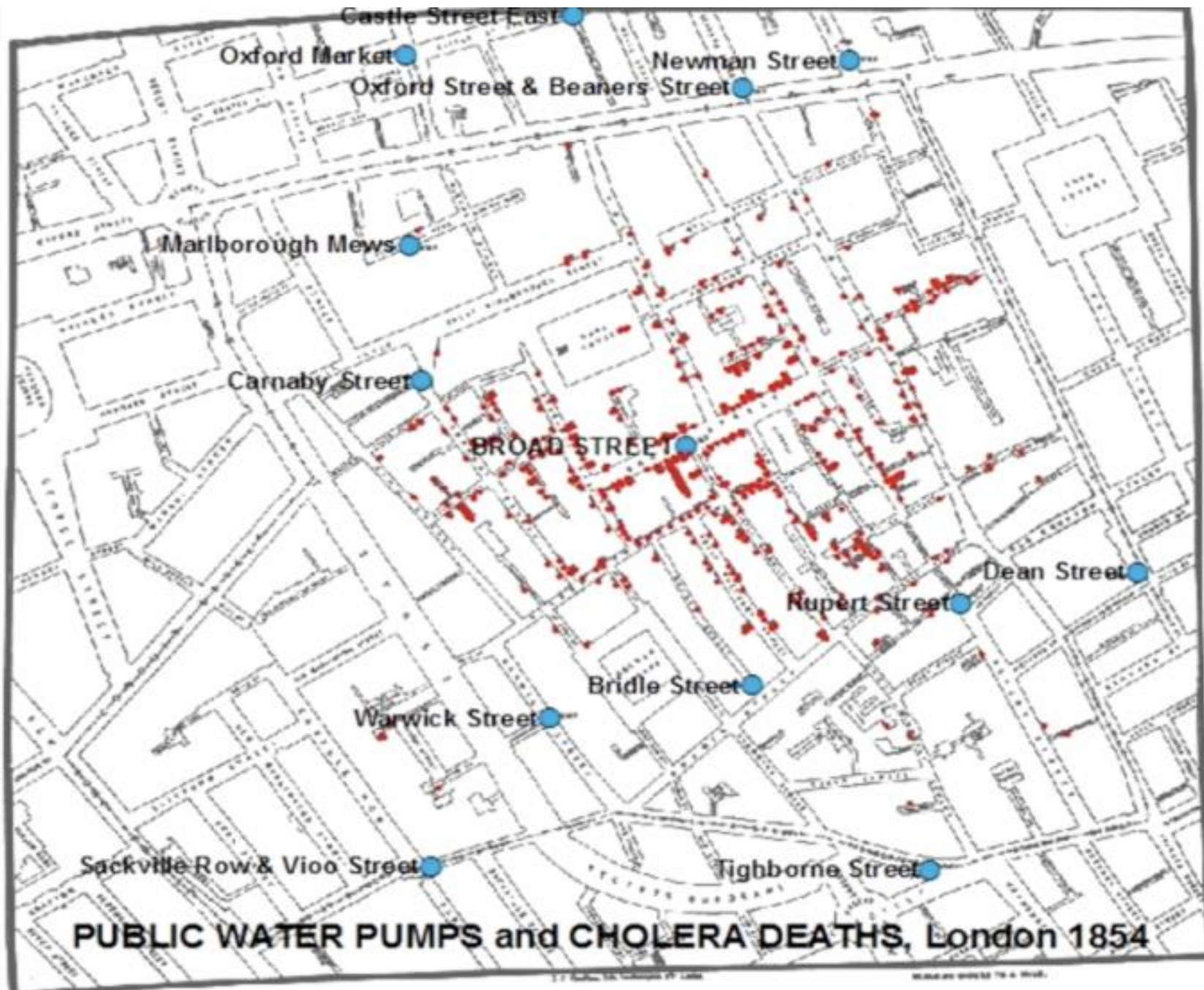
Dr John Snow theorised that cholera reproduced in the human body and was spread through contaminated water.

This contradicted the prevailing theory that diseases were spread by “miasma” in the air.



Dr. John Snow





MAPPING CHOLERA OUTBREAK

SNOW'S WORK

Visualises data on a map.

Shows spatial interconnections between different things.

It allows hypothesising about a possible connection between water quality and cholera spread.

As a result, Snow was successful in arguing for the water pump handle to be removed from the Broad Street pump.



Roger Tomlinson – **the father of GIS**

1960-70

- Directed the development of the Canadian Geographic System (CGIS).
- CGIS was unique because it adopted a layer approach system to map handling.



$$\text{GIS} = \text{G} + \text{IS}$$

Geographic reference + Information system

**Data of spatial coordinates
on the surface of the earth
(Map) – location data**

**Database of attribute data
corresponding to spatial
location and procedures to
provide information for
decision making**

GIS = IS with geographically referenced data

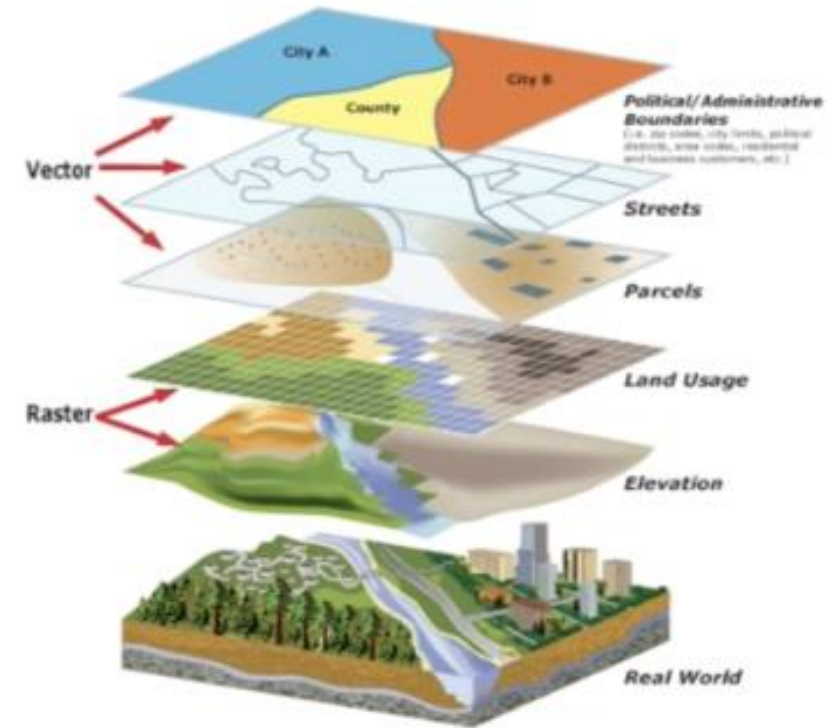
WHAT IS GIS?



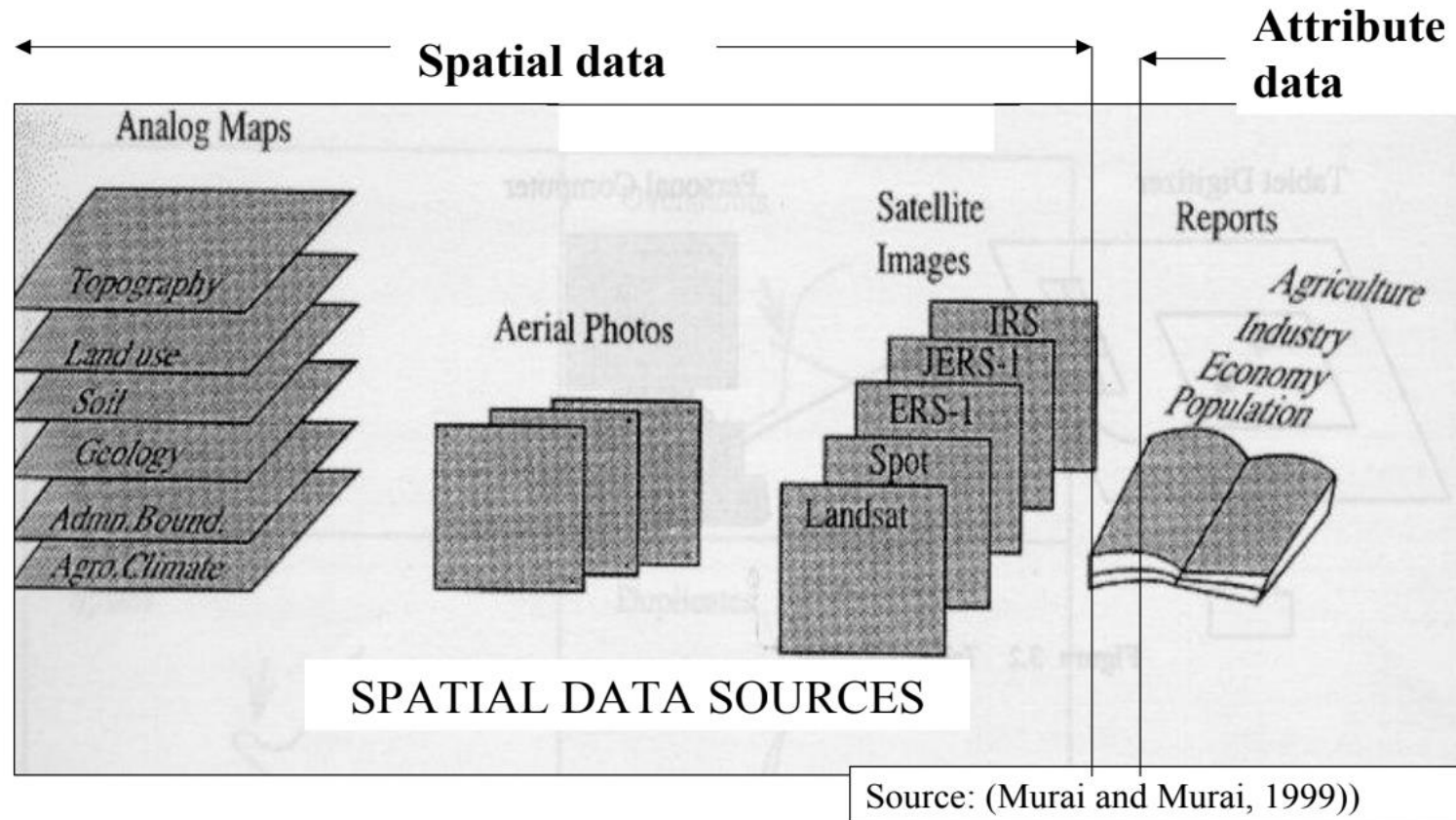
GEOGRAPHIC INFORMATION SYSTEMS (GIS)

A Geographic Information System (GIS) lets us visualize, question, analyse, and interpret data to understand relationships, patterns, and trends.

(Source: ESRI)



Components of Geographic data





WHAT CAN GIS HELP TO ANSWER

- **What exists at a particular location?** Given a geographic reference (e.g. lat, long) for a location, the GIS must describe the features of that location
- **Where can specific features be found?** This is the converse of the first question.
For example, where are the districts with rainfall greater than 500 mm and less than less than 750 mm?
- **Trends or What has changed over time?** This involves answering both questions above.
For example, at what locations are the crop yields showing declining trends?
- **What spatial patterns exist?** If occurrence of a pest is associated with a hypothesized set of conditions of temperature, precipitation, humidity, where do those conditions exist?
- **Modelling or What if ...?** This is a higher-level application of GIS and answers questions like what would be the nitrate distribution in groundwater over the area if fertilizer use is doubled?

GIS is used in multiple disciplines

Education

Journalism

Biology

Ecology

Library Science

Atmospheric sciences

Environmental sciences

Archeology

Economics

Agriculture

Agriculture

Anthropology

Forest science

Geology

History

Political science

Business

Medicine

Geography

Public health

I. Types of GIS Software

- Desktop
- Online/Cloud/Server/Web
 - Mobile

II. Commercial GIS Software

- ArcGIS Pro
- MapInfo Pro
- GeoMedia
- Smallworld
- OpenCities Map

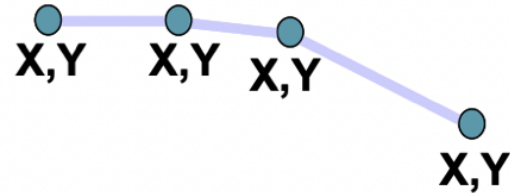
III. Open-Source GIS Software

- QGIS
- gvSIG
- SAGA GIS
- GRASS GIS
- Whitebox GAT

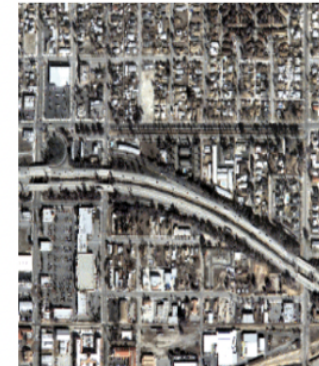
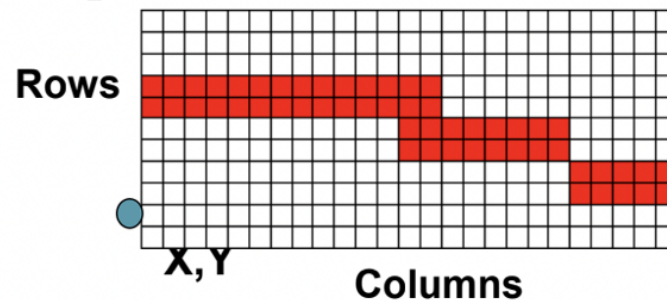
Geospatial Data Representation

Vector and Raster representations

- Vector formats
 - Discrete representations of reality



- Raster formats
 - square cells to model reality



**Reality
(A highway)**

Source: ESRI

COMPUTER REPRESENTATION

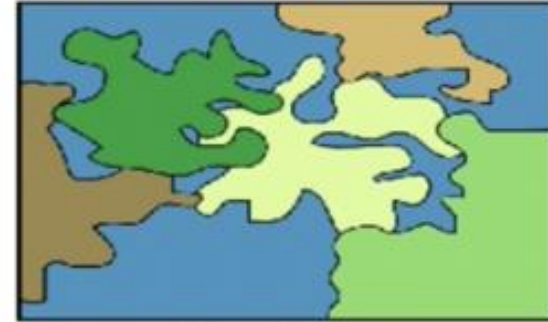
VECTOR MODEL VS RASTER MODEL



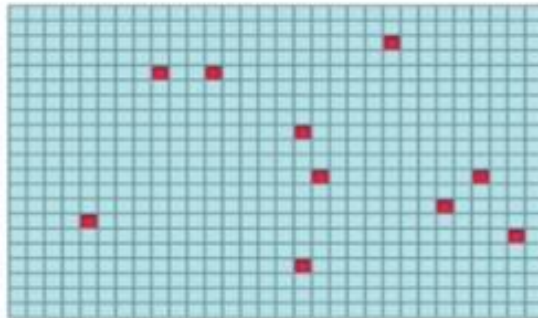
Point features



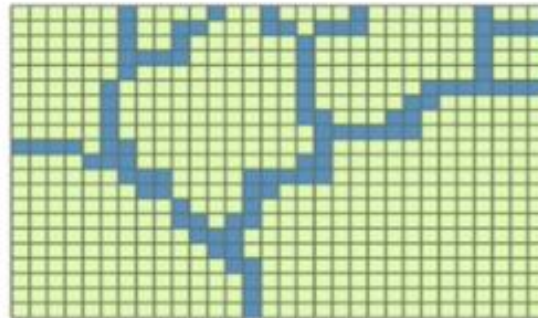
Line features



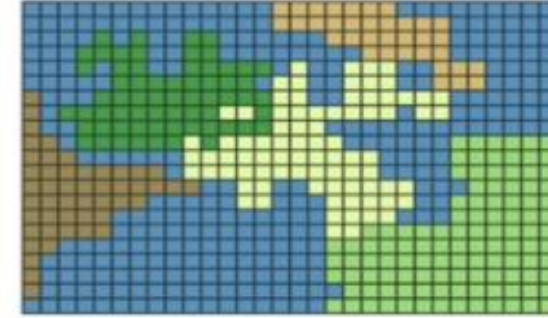
Polygon features



Raster point features



Raster line features



Raster polygon features